

RAJALAKSHMI ENGINEERING COLLEGE

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Course Code	AD23532	Course Name	Principles of Data Science
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Review - 2

Data Preprocessing and Exploratory Data Analysis

Food Waste Prediction and Meal-Planning Optimization for Event Catering

1. Introduction and Scope of Review 2

Review 1 established the problem statement, selected the *Food Wastage Data in Restaurant* dataset (Kaggle, 1,782 records x 11 features) and defined the dual modelling objective: regression on Wastage Food Amount and multi-class classification of booking waste risk.

Review 2 covers the preparation stage that precedes modelling: assessing data quality, cleaning and validating the dataset, transforming and engineering pre-event features, integrating a derived segment-level relation, and conducting exploratory data analysis whose results determine the modelling choices made in Review 3.

All results reported here are produced by the accompanying notebook, `Review2_Preprocessing_EDA.ipynb`, which executes end to end from the raw CSV.

2. Data Quality Assessment

2.1 Structural audit

Property	Result
Records x features	1,782 x 11
Missing values	0 across all 11 columns
Empty or whitespace-padded entries	0
Casing or spelling variants within a category	0
Numeric columns with non-integer values	0
Encoding defect	UTF-8 byte-order mark on the header row

The byte-order mark corrupts the first column name unless the file is read with `encoding='utf-8-sig'`. This is the only formatting defect present.

2.2 Duplicate records

164 rows (9.2% of the dataset) are exact duplicates across all 11 columns. A further 58 rows share an identical feature vector but record a different target value.

Because the dataset contains only eight low-cardinality categorical columns, 61 distinct (guests, quantity) combinations and 26 distinct target values, some exact du-

plication is expected by coincidence between genuinely distinct bookings. Two null models were simulated to quantify the expected coincidence rate:

Null model	Expected duplicate rows
A — all 11 columns treated as independent	0.8
B — correlated block preserved, five weak categorical columns shuffled (300 permutations)	64.7 ± 7.3
Observed	164

Null model B preserves the real association between guest count, quantity, pricing tier and preparation method, and is therefore the appropriate baseline. It indicates that approximately 65 of the 164 duplicates are coincidental collisions between distinct bookings, and approximately 100 are genuine repeated records.

The 58 feature-duplicate rows with differing targets are not errors. They quantify the irreducible noise in the data and are used in Section 6.4.

2.3 Outliers and cross-field validity

Column	1.5 x IQR outliers	3.0 x IQR outliers	$ z > 3$	Skew	Kurtosis
Number of Guests	0	0	0	0.44	-0.51
Quantity of Food	0	0	0	-0.15	-1.10
Wastage Food Amount	9	0	9	0.50	-0.26

The nine flagged target values (54 and 63 units) lie within the 3.0 x IQR fence and represent the high-waste events that the project is designed to predict.

Domain-rule validation confirms that guest count and quantity are strictly positive, waste is non-negative, and waste never exceeds the quantity prepared. Fourteen records show a quantity lower than the guest count (400 guests, 350 units). Since the unit of measurement for Quantity of Food is not documented on the source, and waste ranges from 2.1% to 15.4% of quantity across the dataset — consistent with a weight-based unit — these records are flagged as a unit ambiguity and retained.

2.4 Value granularity

Check	Result
Distinct (guests, quantity) combinations	61 across 1,782 records
Coverage of the 7 most frequent combinations	67.4% of all records
Distinct target values	26, ranging 10–63
Coverage of the 6 most frequent target values	77.2%, all multiples of 5
Target values divisible by 5	77.2% (uniform expectation approximately 20%)

This granularity indicates that the dataset is synthetically generated or template-based rather than collected from operational catering records. The analytical methodology is unaffected, but the finding constrains the interpretation of model performance

and is carried into Section 9.

2.5 Data quality issue register

ID	Issue	Finding	Action
DQ-01	Missing values	None in 11 columns	No imputation required
DQ-02	Duplicate records	164 exact duplicates, approximately 65 coincidental	Remove all; impact quantified
DQ-03	Feature-duplicate conflicts	58 rows, identical features, differing target	Retain; used to bound attainable R^2
DQ-04	Outliers	9 target values with $ z > 3$	Retain; influence tested
DQ-05	Cross-field logic	14 records with quantity < guests	Flag as unit ambiguity; retain
DQ-06	Encoding	UTF-8 BOM on header	Read with utf-8-sig; snake_case rename
DQ-07	Value granularity	61 combinations; 77% round targets	Recorded as a limitation
DQ-08	Class balance	Purchase History imbalanced 7.0x	Note for effect-size interpretation

3. Data Preprocessing and Cleansing

3.1 Formatting and standardisation

Column names were normalised to snake_case and the byte-order mark removed. No categorical recoding was required, as Section 2.1 established that no casing, whitespace or spelling variants exist.

3.2 Deduplication

All 164 exact duplicates were removed, reducing the dataset from 1,782 to 1,618 records (-9.2%). The distributional impact was measured before accepting the step:

Impact measure	Result
Largest change in a numeric column mean	0.32%
Largest categorical share drift	1.03 pp
Kolmogorov-Smirnov tests rejecting equality of distribution	0 of 3

Deduplication is therefore distribution-preserving. Its justification is the prevention of train/test contamination, since a duplicated record can otherwise appear in both partitions:

Configuration	Random Forest test R^2
Duplicates retained; split applied afterwards	0.9257
Duplicates removed before splitting	0.9154
Difference attributable to contamination	+0.0102

This revises the Review-1 figure of $R^2 = 0.926$, which was obtained on a contaminated split. The corrected estimate is $R^2 = 0.9154$ on the held-out test set and 0.904 under 5-fold cross-validation.

3.3 Outlier treatment

The nine high-waste records were retained. Random Forest test R^2 is 0.9154 with them included and 0.9159 with them removed — a difference within cross-validation variance. Removing them would discard the highest-cost operational cases without measurable benefit.

3.4 Data validation

A reusable validation routine enforces six invariants on the cleaned dataset: absence of missing values, absence of duplicates, positive numeric ranges, target within $[0, \text{quantity}]$, no unseen categorical levels, and stable data types. The routine raises an error rather than a warning, and is applied both after cleaning and after every subsequent transformation stage. The cleaned dataset (1,618 x 11) is exported as `out/cleaned_event_catering.csv`.

4. Data Transformation and Feature Engineering

4.1 Partitioning before target-derived statistics

The dataset was partitioned 80/20 (1,294 training and 324 test records, seed 42) before any statistic was derived from the target variable. This corrects a methodological issue carried from Review 1, where the LOW/MEDIUM/HIGH risk tiers were derived from quantiles of the full dataset, allowing test-set values to influence test-set labels. The cut points are now computed on the training partition alone ($Q_1 = 20$, $Q_3 = 35$) and applied unchanged to the test partition. A Kolmogorov-Smirnov test confirms the two partitions share the same target distribution ($p = 0.635$).

4.2 Risk-tier target and its class distribution

296 records take the value 20 and 177 take the value 35 — 29.2% of the dataset concentrated on the two quantile cut points. The rule $y < Q_1$ therefore assigns every record at exactly 20 to MEDIUM, and $y \geq Q_3$ assigns every record at exactly 35 to HIGH:

Risk tier	Training share	Test share
LOW	15.0%	13.6%
MEDIUM	50.9%	50.0%
HIGH	34.2%	36.4%

The quantile rule does not produce a 25/50/25 split on a target with heavy ties. A trivial classifier predicting MEDIUM for every booking would score 50.9% accuracy, which makes accuracy an unsuitable evaluation metric for Review 3.

4.3 Engineered features

All engineered features use information available at booking time:

Feature	Definition	Rationale
<code>food_per_guest</code>	$\text{quantity} / (\text{guests} + 1)$	Per-head preparation intensity
<code>surplus_units</code>	$\text{quantity} - \text{guests}$	Absolute over-preparation
<code>pricing_ordinal</code>	Low = 0, Moderate = 1, High = 2	Preserves the natural ordering of tiers

Feature	Definition	Rationale
is_high_price	1 if pricing = High	Isolates the dominant category
is_sit_down	1 if preparation = Sit-down Dinner	Isolates the highest-waste service style
guest_scale	Binned guest count (4 bands)	Non-linear scale effects

Expected Waste per Guest is excluded from the model inputs because it is derived from the target. It is computed on the Streamlit dashboard after a prediction is produced.

Two automated checks guard the input set: no feature name may contain a target-derived substring, and no numeric input may correlate with the target above $|r| = 0.95$. The highest legitimate correlations are is_high_price (0.765), pricing_ordinal (0.663) and number_of_guests (0.650).

4.4 Encoding and scaling

Pricing is ordinally encoded; the remaining categorical variables are one-hot encoded, giving a 33-column design matrix. StandardScaler is fitted on the training partition only and applied to the linear, SVM and KNN baselines. Tree ensembles use the unscaled matrix, as they are invariant to monotonic feature scaling.

5. Data Integration

The source is a single flat table. A second relation was therefore derived and integrated: an **operational benchmark table** keyed on (pricing tier x preparation method), the two factors identified in Section 6 as dominant. It contains nine segments with the mean, median and standard deviation of waste, mean waste per guest, and a sample-size reliability flag.

Every statistic in the benchmark table is computed on the **training partition only** (1,294 records) and then merged onto both partitions. Computing it on the full dataset would transfer test-set target information into test-set features.

Integration property	Result
Join type	Left join, validated many-to-one
Row-count integrity	1,294 -> 1,294 (train), 324 -> 324 (test); no fan-out
Unmatched keys	0 — every test segment exists in the training benchmark
Segments below the 50-record reliability threshold	0 of 9
Correlation of benchmark mean with target (train / test)	0.812 / 0.801

The close agreement between training and test correlation confirms that no target information leaked through the merge. A leaked target encoding would show a correlation near 1.0 on the training partition and a substantially lower value on the test partition.

Measured effect on model performance. With the merged benchmark columns included, 5-fold cross-validated R^2 changes from 0.9035 to 0.9044, a difference of +0.0009. The benchmark features are a deterministic function of two variables already present in the feature set and therefore add no new information. The integra-

tion is retained for the segment-level views and the dashboard, where per-segment benchmarks are operationally useful, but it is not claimed as a modelling improvement.

Analytical views derived from the integrated dataset:

View	Records	Share	Mean waste
high_risk_bookings	442	34.2%	40.60
premium_sit_down	278	21.5%	41.45
large_events_300plus	801	61.9%	32.97
budget_finger_food	123	9.5%	18.59

premium_sit_down is the operational target segment: it carries the highest mean waste and is where the donation-alert rule will fire most frequently.

6. Exploratory Data Analysis

All exploratory analysis was conducted on the training partition, so that no observation used to inform a modelling decision is drawn from the held-out test data. Each visual result is accompanied by the corresponding statistical test.

6.1 Univariate analysis

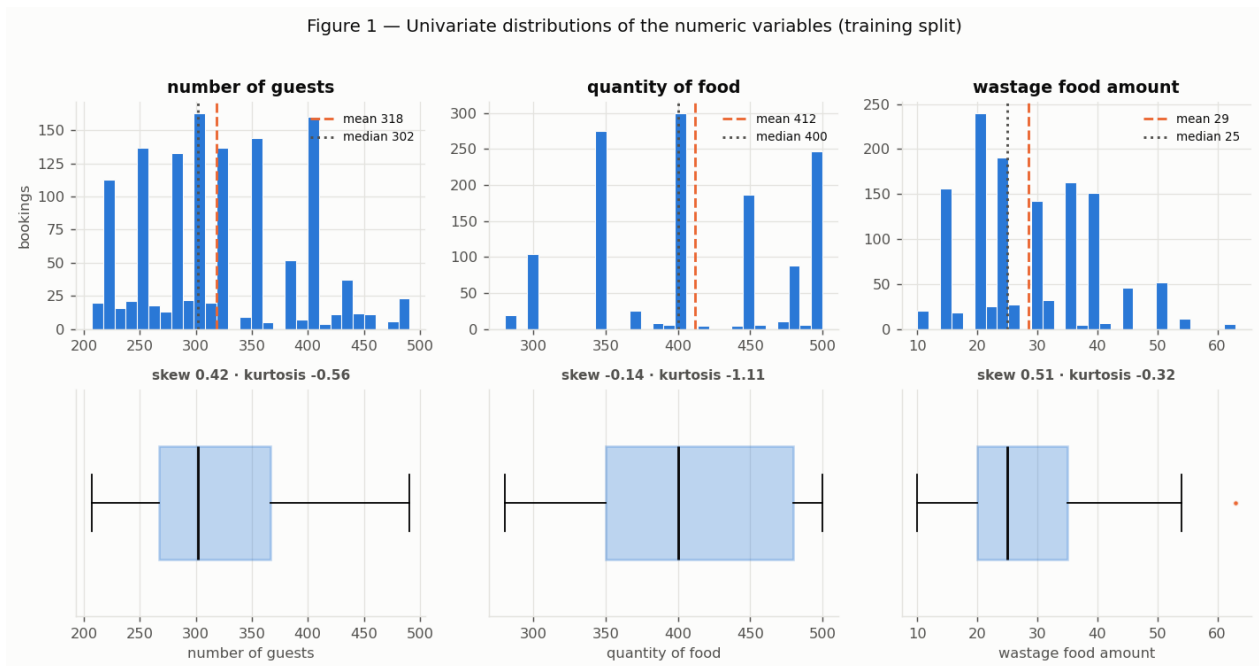


Figure 1 — Distributions of the numeric variables. Guest count is mildly right-skewed (0.44); quantity of food is approximately symmetric but platykurtic (kurtosis -1.10), a consequence of the limited value pool identified in Section 2.4; waste is right-skewed (0.50) with a tail of high-waste events. Shapiro-Wilk rejects normality for all three variables ($p < 10^{-10}$), so non-parametric tests are reported alongside parametric ones throughout.

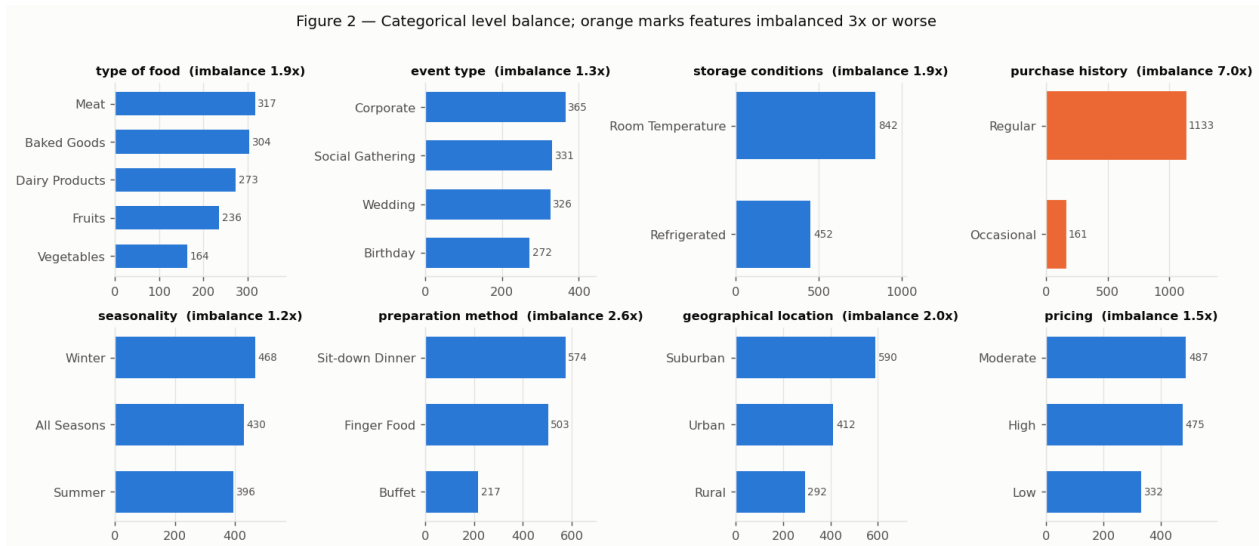


Figure 2 — Categorical level balance. Purchase History is imbalanced 7.0x (1,133 Regular against 161 Occasional bookings) and Preparation Method 2.6x. Effects estimated for a minority level rest on proportionally fewer observations.

6.2 Bivariate analysis

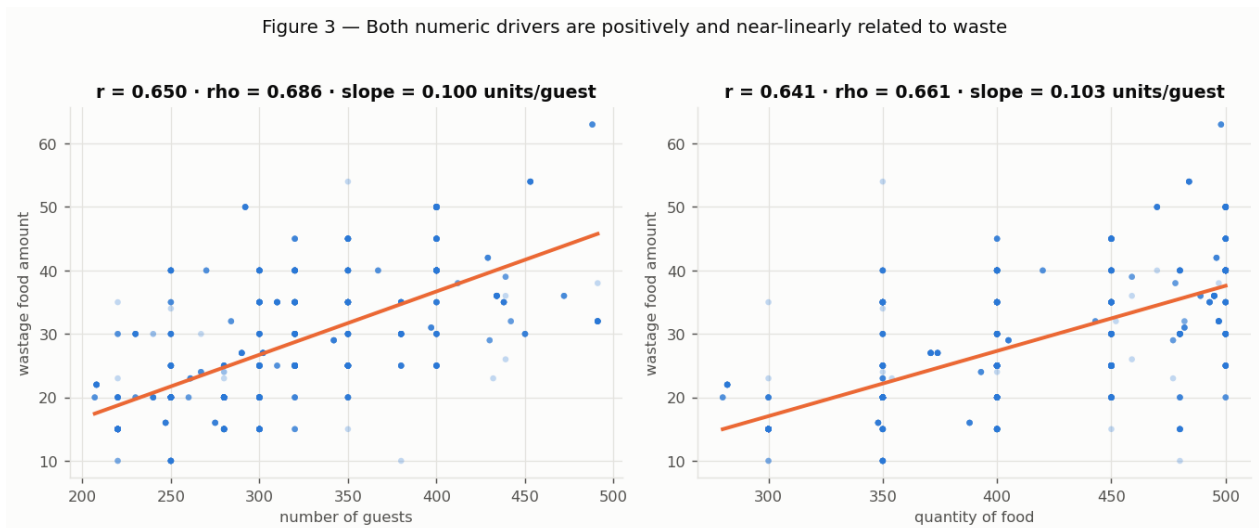


Figure 3 — Numeric drivers against the target. Guest count correlates with waste at $r = 0.650$ ($\rho = 0.686$) and quantity of food at $r = 0.641$ ($\rho = 0.661$). The close agreement between Pearson and Spearman coefficients confirms a monotone, approximately linear relationship.

Each categorical variable was tested against the target using one-way ANOVA, Kruskal-Wallis as non-parametric confirmation, and η^2 as the effect size (k = number of levels):

Feature	k	F	ANOVA p	Kruskal p	η^2	Effect
Pricing	3	910.59	2.1e-247	7.1e-164	0.585	Large
Preparation Method	3	138.89	2.3e-55	7.2e-56	0.177	Large
Geographical Location	3	5.77	3.2e-03	1.8e-03	0.009	Negligible
Type of Food	5	1.19	0.314	0.463	0.004	Negligible

Feature	k	F	ANOVA p	Kruskal p	η^2	Effect
Seasonality	3	1.62	0.197	0.176	0.003	Negligible
Event Type	4	0.51	0.676	0.615	0.001	Negligible
Purchase History	2	0.82	0.366	0.437	0.001	Negligible
Storage Conditions	2	0.08	0.784	0.599	0.000	Negligible

Five of the eight categorical features show no statistically detectable relationship with food waste. Event Type, around which the project is framed, is one of them; Storage Conditions, which domain reasoning suggests should be influential, has $\eta^2 = 0.0001$. Both results are consistent with the synthetic data structure identified in Section 2.4.

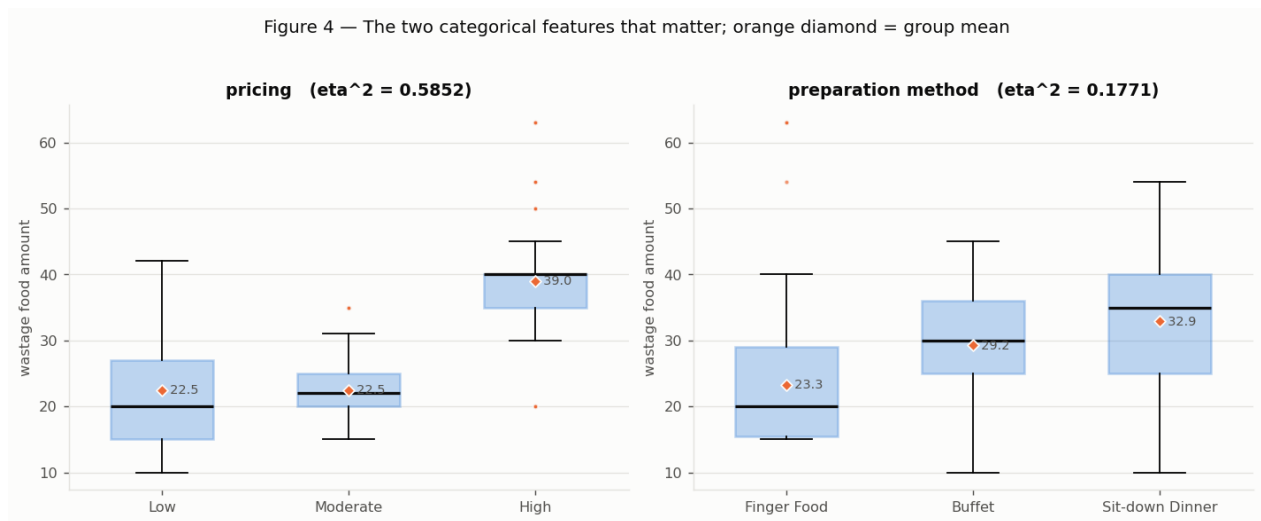


Figure 4 — The two categorical features with a measurable effect. High-tier bookings average 39.0 units of waste against 22.5 for Low-tier and 22.5 for Moderate-tier bookings. Sit-down Dinner service averages 32.9 units against 29.2 for Buffet and 23.3 for Finger Food.

6.3 Multivariate analysis

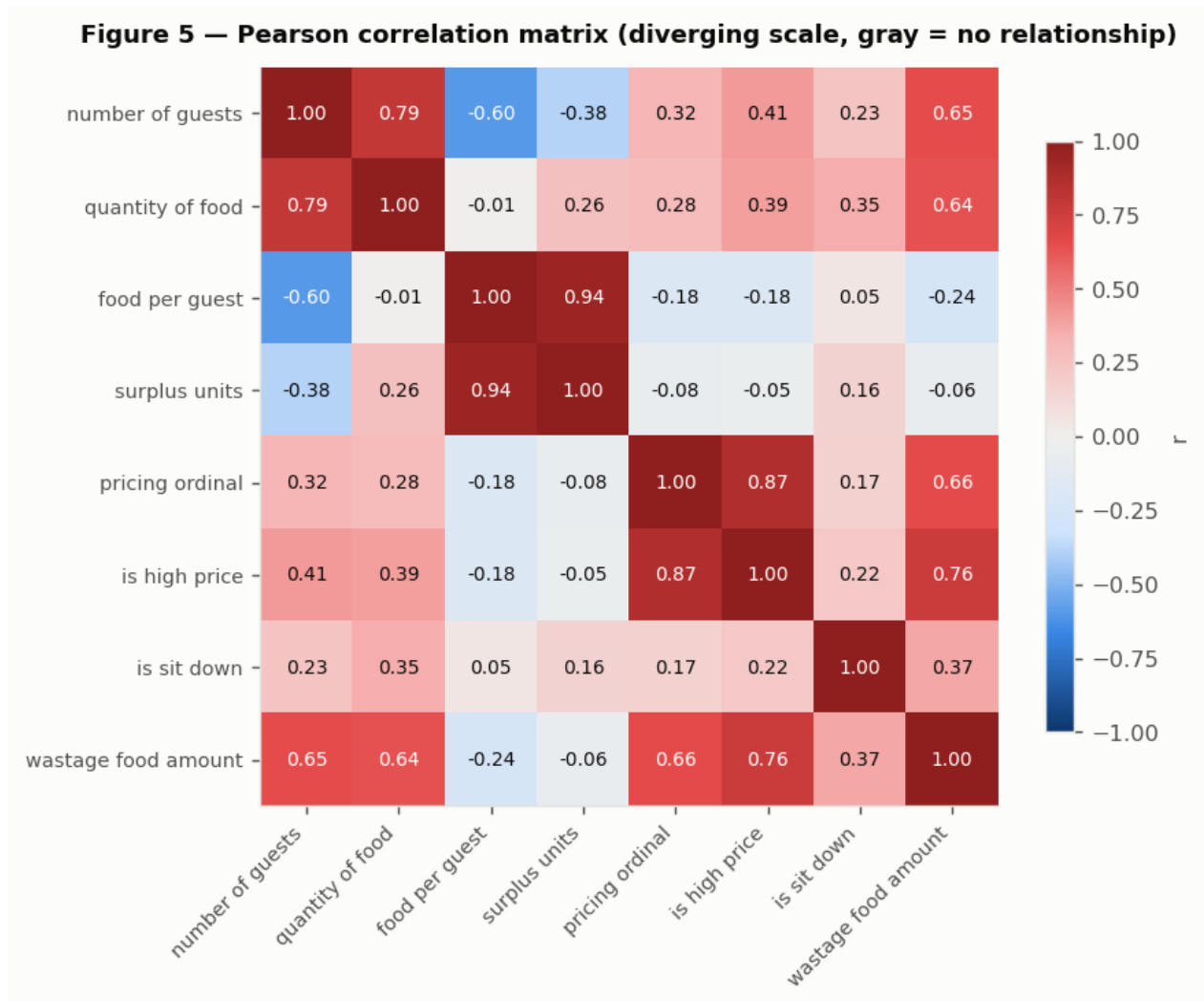


Figure 5 — Pearson correlation matrix. Three feature pairs exceed $|r| = 0.7$:

Pair	r	VIF	Interpretation
food_per_guest + surplus_units	0.938	8.37	Exceeds the VIF > 5 threshold; two encodings of the same quantity. A linear model must include one, not both
pricing_ordinal + is_high_price	0.866	4.01	Intentional redundancy; a tree uses the ordering and the single split point differently
number_of_guests + quantity_of_food	0.791	2.67	Acceptable, but coefficients on these two cannot be read as independent effects

Tree ensembles are unaffected by collinearity, which also explains why impurity importance distributes credit across correlated columns in Figure 9.

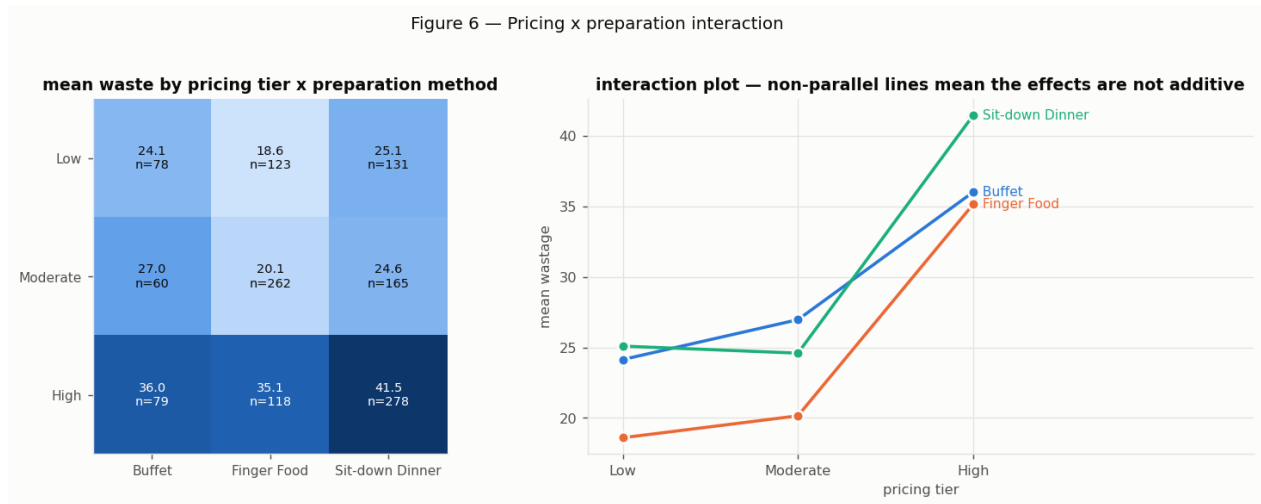


Figure 6 — Pricing tier by preparation method. A saturated cell-means model tested against the additive model gives $F(4, 1285) = 11.54$, $p = 3.3 \times 10^{-9}$: the interaction is statistically significant. It manifests as a rank reversal:

Pricing tier	Ranking of preparation methods by mean waste
Low	Sit-down Dinner (25.1) > Buffet (24.1) > Finger Food (18.6)
Moderate	Buffet (27.0) > Sit-down Dinner (24.6) > Finger Food (20.1)
High	Sit-down Dinner (41.5) > Buffet (36.0) > Finger Food (35.1)

Buffet service ranks above Sit-down Dinner in the Moderate tier and below it in the other two. An additive model cannot represent this ordering; a tree-based model can.

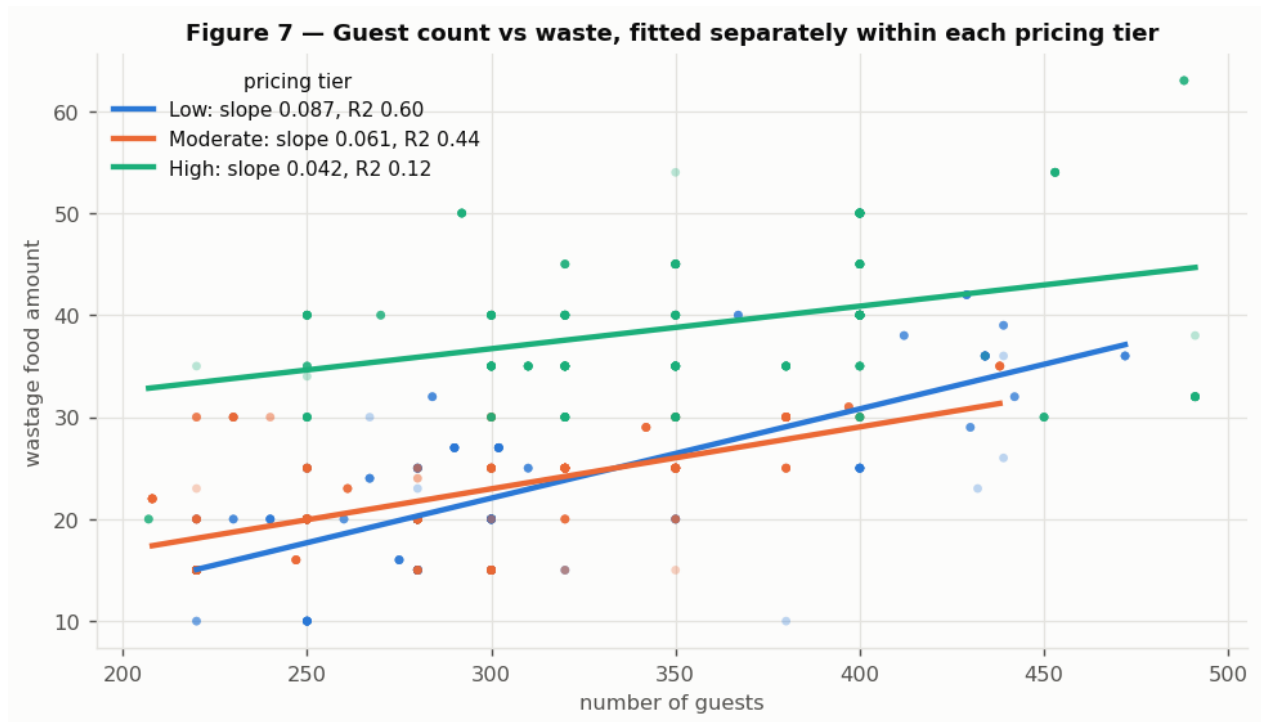


Figure 7 — Guest count against waste within each pricing tier. The regression slope is 0.087 units per guest in the Low tier, 0.061 in the Moderate tier and 0.042

in the High tier — a factor of 2.1 — while within-tier explanatory power falls from $R^2 = 0.60$ to 0.12. A single global linear coefficient for guest count is therefore misspecified.

6.4 Attainable predictive performance

Records sharing an identical feature vector but recording different target values (DQ-03) establish an upper bound on attainable accuracy, since no model can produce two different predictions from one input. Pooling the within-group variance of such records gives:

Conditioning set	Degrees of freedom	Residual variance	Implied R^2 ceiling
All 10 features, post-deduplication	58	26.85	0.752
The 4 features with a measurable effect	1,502	7.26	0.933

The first estimate is not usable. Only 55 replicate groups survive deduplication, and they survive because their target values differed — groups whose values coincided were removed as exact duplicates. This is selection on the outcome, and it overstates the residual variance. The second estimate conditions on the four features shown in Section 6.2 to carry all the signal, rests on 1,502 degrees of freedom, and is not subject to that selection. $R^2 \approx 0.933$ is therefore adopted as the attainable ceiling.

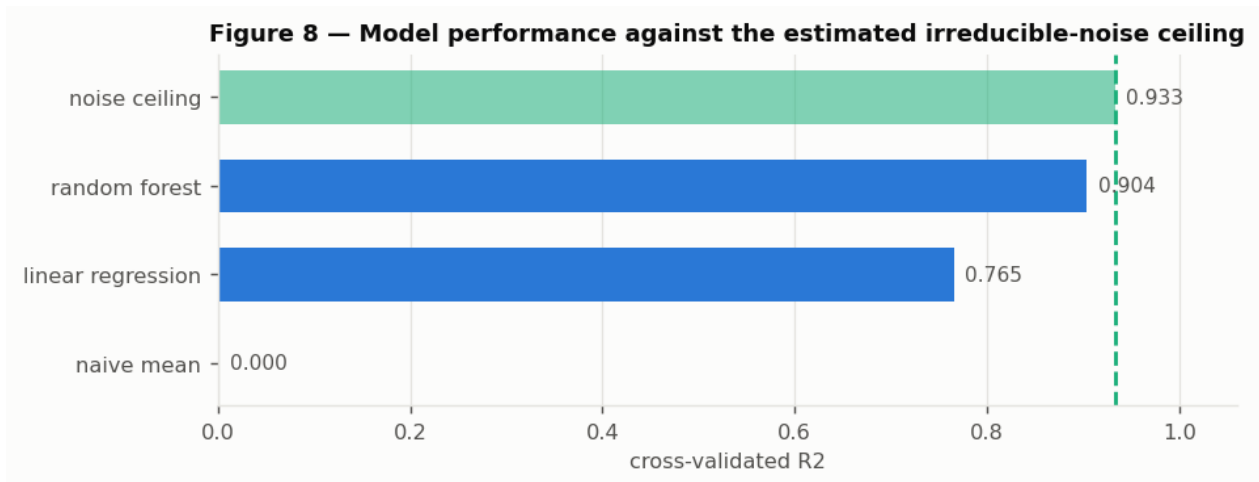


Figure 8 — Model performance against the estimated ceiling.

Model	5-fold cross-validated R^2
Naive mean baseline	0.000
Linear Regression	0.765
Random Forest	0.904
Estimated noise ceiling	0.933

The Random Forest has captured approximately 97% of the learnable signal, leaving 0.029 of genuine head-room. A reported R^2 materially above 0.95 would indicate leakage rather than improvement.

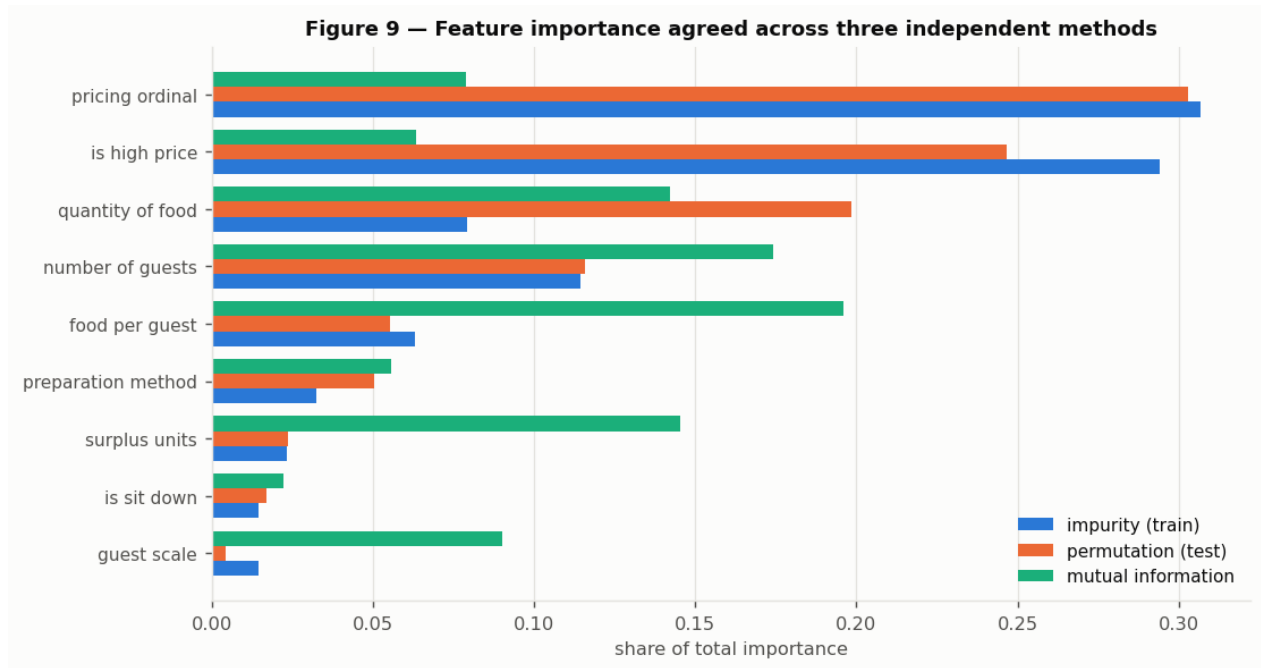


Figure 9 — Feature importance under three methods. Impurity importance (training), permutation importance (test) and mutual information were computed independently, since impurity importance alone is known to be biased. All three rank the pricing variables, quantity of food and guest count above the remainder. Five features record negative permutation importance, indicating that shuffling them improves test predictions.

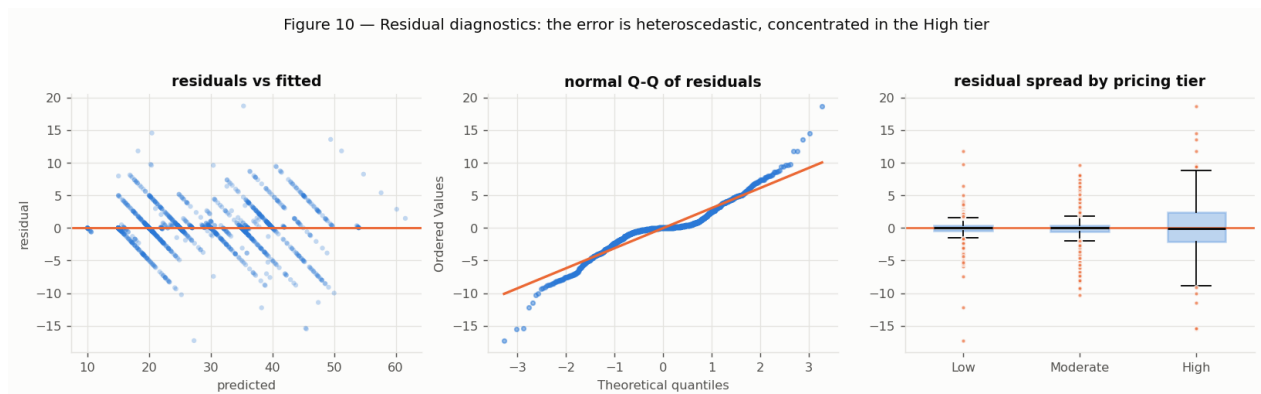


Figure 10 — Residual diagnostics. Residual standard deviation is 2.18 in the Low tier, 2.76 in the Moderate tier and 4.15 in the High tier: the error is heteroscedastic and concentrated in the segment carrying the greatest operational cost. The Q-Q plot shows heavier-than-normal tails at both ends. The diagonal banding in the left panel arises because the target takes only 26 discrete values, so each actual value traces a line of slope -1 in a residual-versus-fitted plot.

7. Key Insights and Interpretation

ID	Insight	Supporting evidence
I-1	Pricing tier is the dominant driver of waste	$\eta^2 = 0.585$; High tier 39.0 against Low 22.5
I-2	Only 4 of 10 features carry signal	Five features with $\eta^2 < 0.01$ and negative permutation importance

ID	Insight	Supporting evidence
I-3	Pricing and preparation method interact	$F(4, 1285) = 11.54$, $p = 3.3e-9$; rank reversal across tiers
I-4	The guest-count effect is tier-dependent	Slope 0.087 (Low) against 0.042 (High)
I-5	Error variance is not constant	Residual SD 2.18 to 4.15 across tiers
I-6	Attainable accuracy is bounded	Identical inputs with differing targets imply $R^2 \leq 0.933$
I-7	Duplicates were a source of leakage	R^2 0.9257 contaminated against 0.9154 clean
I-8	The dataset is synthetic or templated	61 (guest, quantity) pairs; 77.2% of targets are multiples of 5
I-9	Risk tiers are not evenly distributed	15.0 / 50.9 / 34.2 rather than 25 / 50 / 25
I-10	Integrated benchmark features add no predictive value	Cross-validated R^2 change of +0.0009

7.1 Verification of the reduced feature set

Insight I-2 was tested rather than adopted. A model restricted to the four signal-carrying source variables was compared with the full 33-column matrix over identical folds:

Design	5-fold cross-validated R^2
Full design matrix (33 columns)	0.9035 ± 0.0167
Reduced feature set (4 source variables)	0.9066 ± 0.0199
Paired t -test across folds	$t = -0.52$, $p = 0.633$

The two designs are statistically indistinguishable. The reduced model is therefore preferred: it is simpler, faster to train, and more readily interpretable by a catering operator.

7.2 Implications for model selection in Review 3

Question addressed by EDA	Evidence	Consequence for model selection
Are the relationships linear?	Slope varies by a factor of two across tiers (Fig. 7)	Linear Regression is retained as an interpretable baseline only
Are there interactions between features?	$F = 11.54$; rank reversal (Fig. 6)	Random Forest and XGBoost are the appropriate model family
Is feature scaling required?	Tree models are scale-invariant	Scaler fitted on train, applied only to linear, SVM and KNN baselines
How many features should be used?	4 of 10 carry signal (Fig. 9); reduced set matches full set	Adopt the reduced feature set
Is the error homoscedastic?	Residual SD doubles across tiers (Fig. 10)	Report per-tier MAE and RMSE; produce prediction intervals

Question addressed by EDA	Evidence	Consequence for model selection
What performance target is realistic?	Noise ceiling of 0.933 (Sec. 6.4)	Target 0.933; treat higher values as evidence of leakage
Are the risk classes balanced?	15.0 / 50.9 / 34.2 distribution	Evaluate with macro F1 and per-class recall, not accuracy

8. Updated Project Documentation

8.1 Preprocessing and analysis workflow

Raw CSV (1,782 x 11)

- > Read with utf-8-sig; normalise column names [DQ-06]
- > Data quality assessment: missing values, duplicates, outliers, cross-field validity, value granularity [Section 2]
- > Remove 164 exact duplicates; measure distributional impact [1,618 x 11]
- > Validation contract: six invariants, error on failure [Section 3.4]
- > Train / test partition (80/20, seed 42)
- > Derive risk tiers from training quantiles only [Section 4.2]
- > Engineer six pre-event features [Section 4.3]
- > Leakage checks: name rule and correlation rule [Section 4.3]
- > Ordinal and one-hot encoding -> 33-column matrix; StandardScaler fitted on training partition only [Section 4.4]
- > Build benchmark relation from training partition; merge [Section 5]
- > Exploratory data analysis on the training partition [Section 6]
- > Pipeline validation checks [Section 8.3]

8.2 Artefacts produced

File	Contents
Review2_Preprocessing_EDA.ipynb	Reproducible notebook with all code and outputs
out/cleaned_event_catering.csv	Cleaned analytical dataset (1,618 x 11)
out/segment_benchmark_table.csv	Integrated benchmark relation (9 segments)
out/data_quality_register.csv	Issue register DQ-01 to DQ-08
out/categorical_effect_sizes.csv	ANOVA, Kruskal-Wallis and η^2 results
out/feature_importance.csv	Three-method feature importance comparison
out/key_insights.csv	Insights I-1 to I-10 with supporting evidence
out/view_*.csv	Four analytical subsets
figures/fig01-fig10.png	Complete EDA figure gallery

8.3 Pipeline validation

The notebook concludes with an assertion-based validation stage covering dataset shape and integrity, non-overlap of the training and test partitions, identical design-matrix geometry across partitions, scaler provenance, one-hot encoding consistency, four separate leakage conditions, benchmark-table provenance, join integrity, and the presence of every exported artefact. All checks pass, and the stage raises an error rather than a warning so that a defective pipeline cannot be exported silently.

9. Limitations

1. The dataset is synthetically generated or template-based (DQ-07). Results demonstrate the analytical methodology; they are not evidence of achievable accuracy on operational catering records.
2. Deduplication removes an estimated 65 legitimate distinct bookings that cannot be distinguished from genuine repeated records.
3. No date field exists, so weekday, seasonal, holiday and calendar effects cannot be modelled.
4. Pricing is recorded as a three-level category rather than a continuous value, restricting price analysis to tier-level comparison.
5. The attainable-performance ceiling is a variance estimate derived from grouped records and shifts with the conditioning set; it bounds expectations rather than defining a constant.
6. Reported test performance derives from a single partition. Review 3 should report repeated cross-validation.
7. Model outputs must inform preparation decisions rather than automatically reduce food quantities, as guest-contract minimums, dietary requirements, service buffers and food-safety rules lie outside the scope of the model.

10. Evaluation Criteria Mapping

Criteria	Marks	Sections addressing it
Data quality assessment and identification of data issues	3	Section 2; register DQ-01 to DQ-08
Data preprocessing and cleansing techniques	5	Section 3; measured impact of each step; validation contract
Data transformation, feature engineering, and data integration	5	Sections 4 and 5; leakage checks; training-only benchmark merge
Exploratory Data Analysis using appropriate statistical methods and visualizations	10	Section 6; Figures 1-10; ANOVA, Kruskal-Wallis, η^2 , interaction F-test
Interpretation of insights and justification for future model selection	4	Section 7; insights I-1 to I-10; model-selection table
Technical report	3	This document and the accompanying notebook
Total	30	